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	SECOND TERM SUMMATIVE ASSESSMENT 2025–26 **STD X – CHEMISTRY (EM) – ANSWER KEY** **Section A (1 × 4 = 4)**	
Sl. No	**Answer Indicators**	**Score**
1	C. H_2, Cl_2 and NaOH	1
2	B. Statement 1 is correct but Statement 2 is incorrect	1
3	Matching: **a–iv, b–iii, c–i**	1
4	Both Assertion and Reason are correct and Reason explains Assertion	1

Section B (2 × 7 = 14)

Sl. No	**Answer Indicators**	**Score**
5	a.Pentanal – $CH_3–CH_2–CH_2–CH_2–CHO$	1
	b. Pent-2-yne – $CH_3–C≡C–CH_2–CH_3$	1
6	(a). f-block elements	1
	(b). Any one property of f-block elements ◆ Inner transition ◆ strong magnets ◆ variable oxidation state	1
7	(a) Orbitals identified: 3d and 4p (b) 4p orbital has higher energy since when (n+l) is same, orbital with higher n has higher energy	1 1
8	(a) Fuel cell: converts chemical energy directly into electrical energy	1

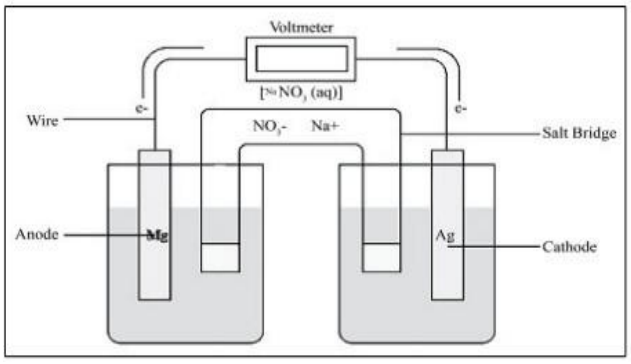
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Sl. No	Answer Indicators	Score
	(b)Any one advantage ◆ High efficiency ◆ Eco-friendly	1
9	(A) (a) . Position isomers: $CH_3–CH_2–CH_2–CH_2OH$ and $CH_3–CH_2–CH(OH)–CH_3$ (b) Functional isomer: $CH_3–CH_2–O–CH_2–CH_3$ OR 9(B) (i) Pentan-2-one – $CH_3–CH_2–CO–CH_2–CH_3$ (ii) Pentan-3-one – $CH_3–CO–CH_2–CH_2–CH_3$	1 1
10	(a) Colour of solution turns blue (b) Copper is oxidised	1 1
11	Any four qualities of an ore ◆ Easy availability ◆ High metal content ◆ Easy separation ◆ Low extraction cost	2

Section C (3 × 6 = 18)

Sl. No	Answer Indicators	Score
12(A)	(a) Ethyne → Ethene: $C_2H_2 + H_2 \rightarrow C_2H_4$	1
	(b) Polythene formation: $n(CH_2=CH_2) \rightarrow (–CH_2–CH_2–)n$	1
	(c) Ethene → Chloroethane: $CH_2=CH_2 + HCl \rightarrow CH_3–CH_2Cl$ OR	1
12(B)	(a) Condensation polymerisation – polymer formation with	1

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Sl. No	Answer Indicators	Score
	elimination of small molecules (b) Example: Nylon-6,6; Monomers – adipic acid and hexamethylene diamine	 2
13	(a) Given: V_1 = 10 L, T_1 = 77°C = 350 K; T_2 = 7°C = 280 K Using Charles' law: $V_1 / T_1 = V_2 / T_2 V_2 = (10 \times 280) / 350 =$ **8 L** (b)Charles	2 1
14	(a)Subshell configuration of M: $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^1\ 3d^{10}$ (b) M^+ : $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^{10}$ M^{2+} : $1s^2\ 2^{s2}\ 2p^6\ 3s^2\ 3p^6\ 3d^9$	1 2
15	(a) Same period elements: B and D (b) Element B – Group 7, Period 4 (c)Element in same group as A: **C**	1 1 1
16(A) 16(B)	(a) Mg–Cu cell gives higher voltage due to larger electrode potential difference (b) Direction of electron flow: Mg → Cu (c) Anode reaction: $Zn \rightarrow Zn^{2+} + 2e^-$ OR  b) Mg--->Ag	1 1 1
17	a) B b) At constant temperature and pressure, volume is directly proportional to number of moles	1 1 1

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Sl. No	Answer Indicators	Score
	c) $V_1 / n_1 = V_2 / n_2$ $V_2 = (640 \times 5) / 20$ $V_2 = 160$ L	

Section D (4 × 1 = 4)

18 A	(a) Molar mass of CO_2 = 44 g mol^{-1}	1
	(b)Volume of 880 g CO_2 at STP = 448 L	1
	(c)Number of molecules = $10 \times 6.022 \times 10^{23}$	1
	(d) M(a) Molar mass of CO_2 = 44 g mol^{-1}	1
	(b)Volume of 880 g CO_2 at STP = 448 L	
	(c)Number of molecules = $10 \times 6.022 \times 10^{23}$	
	(d) 12g C +32g O_2 → 44g CO_2,	
	88 g CO_2 formed is formed from	
	= 12/44 × 88 g C	
	= 24g C	
	Mass of carbon required =24	
18B	OR	
	(a) $N_2+3H_2 \rightarrow 2NH_3$	
	(b) 56g	
	(c) N_2 required for 30 mol H_2 to react is N_2 =30/3 =10 mol	
	Total Nitrogen =20mol , So Remaining N_2 = 20-10=10 mol	
	PREPARED BY RAVI P;HST PHY.SCIENCE; HS PERINGODE, PALAKKAD	